Test Diverse Users

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

TEACHER GUIDE · 1 OF 9

Test Diverse Users

> Prepare Test Diverse Users: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Evaluate across varied cases

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Build the Recommender

Without opening the old lesson, explain one key idea from Build the Recommender. Give an example and a mistake that the idea helps you avoid.

Look for: Implement a deterministic recommender

Reference example, not the only acceptable wording: Both score 8 and qualify; A is alphabetically first under the tie rule.

Permissions and Human Review

Without opening the old lesson, explain one key idea from Permissions and Human Review. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of least privilege and tool scopes

Reference example, not the only acceptable wording: Teacher explanation: Least privilege grants only capabilities required by the task. Tool access should include resource and operation scope, not merely a broad tool name.

A reporting assistant may lookup public rooms but may not write records, send messages or read teacher notes. Check actor, operation and resource before executing a proposed call.

Prompt Experiments

Without opening the old lesson, explain one key idea from Prompt Experiments. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of controlled prompt experiments

Reference example, not the only acceptable wording: Teacher explanation: A prompt experiment compares complete specifications using the same task cases.

Define success before inspecting **output**s; change one intended instruction and preserve other settings. Evaluate both prompts on paired cases with fixed model/version, source records, **output** contract and scoring rubric.

Use development results for selection, then reserve new final cases. Score each case rather than counting attractive sentences.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

The recommender succeeds on 9/10 mystery requests and 2/5 poetry requests. Compute total and separate rates. A team reports only 11/15. Explain what that hides.

Concept to explain

Evaluation must include ordinary, edge, and **underserved** cases.

Term	Meaning
test profile	A description of the kinds of cases and conditions included in an evaluation.
edge case	An unusual, boundary or difficult input that can reveal a failure hidden by routine examples.
underserved	Receiving less adequate support or performance than the intended service requires.
fairness concern	A reason to examine whether a system creates unjustified differences in treatment or outcomes.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach Week 34 as a two-session investigation using fictional data. A fictional chooser passes 5 of 6 development checks but fails its only tie case. An overall **pass rate** can hide that concentrated weakness.

Evaluate ordinary, boundary, invalid, empty and tie cases separately, using requirements rather than duplicated versions of one easy example.

Assess the three stated outcomes through the independent task, not by copying the worked result.

Readiness answer

No; they have become part of development **evidence**.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional **input** records from this lesson; no real student identifiers

End goal for the pictorial case

By the end, I can evaluate varied case types and prioritise an important failure slice.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

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Explain the mechanism

> Model Test Diverse Users using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Profiles: typical reader; very short-book need; bilingual preference; screen-reader format need.

2. Run the same frozen recommender for all four.

3. Record expected, actual, score **evidence**, and issue type.

4. Do not average away one group’s repeated failure.

Recalculate, retrace, or restate the reasoning in your own representation:

> CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

> ### The teaching case
>
> A recommender satisfies 5/6 adventure requests and 3/6 poetry requests.

> ### The teaching case
>
> Compare and investigate.

Exact explanation to model

> ### Why the result follows
>
> Rates are 5/6 and 3/6. Inspect catalogue **coverage**, labels and scoring; twelve tests do not establish cause.

Explain the added worked example

Explain the added worked example · 1

A fictional chooser passes 5 of 6 development checks but fails its only tie case. An overall **pass rate** can hide that concentrated weakness.

Evaluate ordinary, boundary, invalid, empty and tie cases separately, using requirements rather than duplicated versions of one easy example.

The script reports 5 and 6. About 83.3 percent of these checks pass, but that is not an accuracy estimate for all future use.

The cases were deliberately chosen, their frequencies are artificial, and each category has few cases.

Key terms

Term	Meaning
Evaluation set	Cases used to assess a specified version
Failure category	A type of scenario where an expected result is missed
Pass rate	Passing cases divided by executed cases
Coverage	The variety of requirements and conditions tested

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

> **If ten copies of one passing ordinary case are added, what becomes the pass fraction? Does coverage improve?**

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Test Diverse Users

A fictional recommender is tested on distinct ordinary, equality, invalid-**input** and tie cases.

The facts we will use

> **Supplied facts**
>
> Recorded full passes: ordinary 4/4; equality 1/3; invalid 2/3; ties 1/2.

> **Supplied facts**
>
> Categories are disjoint and every case was run on the same frozen version with the same contract.

AI ACADEMY / AI-2/34

Disjoint evaluation categories

	Full passes	Cases run
Ordinary	4	4
Equality	1	3
Invalid	2	3
Ties	1	2

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/34

Test Diverse Users

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Total count

Passes 4+1+2+1=8; cases 4+3+3+2=12.

2. Overall rate

8/12≈66.7% full passes.

3. Find weak slices

Equality 1/3; invalid 2/3; ties 1/2; ordinary 4/4.

4. Prioritise repair

Inspect failed equality traces and any invalid-input safety consequence, not only the total.

CHANGE ONE THING

Someone reports only the ordinary slice.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Total count	Passes 4+1+2+1=8; cases 4+3+3+2=12.
2. Overall rate	8/12≈66.7% full passes.
3. Find weak slices	Equality 1/3; invalid 2/3; ties 1/2; ordinary 4/4.
4. Prioritise repair	Inspect failed equality traces and any invalid-input safety consequence, not only the total.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Varied tests expose behavior an ordinary demonstration misses. The overall result combines disjoint cases once each; overlapping labels would require unique-case accounting.

Prioritisation also depends on consequences: a malformed **input** that triggers an unauthorised action can be more serious than a harmless tie-order failure. Retain each failed case and its expected **output**.

What this example does not establish

Keep the stated limit

Twelve designed cases describe this test suite, not an unbiased estimate of future usage accuracy.

Change one thing in the pictured case

Change one thing

Change: Someone reports only the ordinary slice.

Predicted result

Prediction: The claim becomes 4/4 on ordinary cases, not 100% across all tests.

Why the result follows

Reason: Eight nonordinary cases and their failures have been omitted.

Ask while showing the picture

Ask while showing the picture · 2

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Test Diverse Users, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: A recommender satisfies 5/6 adventure requests and 3/6 poetry requests.

2. State the decision to explain

Write the question to explain: Compare and investigate.

3. Reveal the worked reasoning

Reveal after **prediction**s: Rates are 5/6 and 3/6. Inspect catalogue **coverage**, labels and scoring; twelve tests do not establish cause.

Hold back the independent task

Keep for an independent attempt: A recommender serves 6 of 6 easy-book requests but 1 of 4 advanced-book requests. Report both results and identify the next test priority.

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant

details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

15/16 pass, but no new condition was tested. The larger fraction is misleading as **evidence** of broader quality. Exit: The old case is now a regression/development case.

Document the fix and use fresh reserved cases for a new independent final check.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.

Stage	Teacher action
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Completion practice

Completion task: Name five distinct categories for the chooser evaluation.

Reasoning to listen for

Teacher check: Ordinary selection, duration boundaries, invalid records, empty/no-match **input** and ties/order reversal.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Evaluating varied cases. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: For a rule-based chooser, evaluate conformance to its policy separately from whether users find suggestions useful. Neither a pass fraction nor an

> invented satisfaction score proves educational benefit.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

> Can these records establish performance for different user groups?

Reasoned teacher answer

No. They describe **input**-case types. No user-group membership or group-specific outcomes were supplied.

Changed-case reasoning: The claim becomes 4/4 on ordinary cases, not 100% across all tests. Eight nonordinary cases and their failures have been omitted.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Test Diverse Users investigation and use the first unsupported step to choose a teaching response.

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Create four meaningfully different profiles, predict outcomes, run, classify failures, and compare results by profile rather than only overall.

> What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A successful run proves the conclusion: For a rule-based chooser, evaluate conformance to its policy separately from whether users find suggestions useful.

Neither a pass fraction nor an invented satisfaction score proves educational benefit.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an explanation. Fresh retry: ordinary 2/2, empty 0/1, boundary 2/2 gives 4/5 overall; the empty case still fails.

Require the failure record and a fresh-case plan.

A convenient shortcut preserves meaning: Hiding the tie failure in an average conceals an unmet requirement. Preserve the failed **input**, expected result, actual result and version.

If you repair using it, classify it as development/regression **evidence** and reserve new final cases.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Test Diverse Users questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Rates are 5/6 and 3/6. Inspect catalogue **coverage**, labels and scoring; twelve tests do not establish cause.

Independent question

Independent question · withhold the key

A recommender serves 6 of 6 easy-book requests but 1 of 4 advanced-book requests. Report both results and identify the next test priority.

Expected independent answer

Teacher key · Expected independent answer

Easy requests: 6/6; advanced: 1/4. Investigate advanced requests and inspect their failures. The combined 7/10 hides this difference.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support

Criterion	0 points	1 point	2 points
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A single overall score describes every user's experience.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

Overall is about 73.3%; mystery is 90%; poetry is 40%. The total hides weaker poetry performance.

Investigate catalogue **coverage**, labels and scoring without assuming the rates identify the cause; small denominators limit conclusions.

Explanations for the additional workbook examples

W1 Read denominators

Workbook question · W1

A report says 8 of 10 tests pass, including 0 of 2 invalid-**input** cases. What is the important weakness?

Teacher reasoning · W1

Invalid-**input** handling fails both tested cases even though the overall pass fraction is 80 percent.

W2 Plan coverage

Workbook question · W2

Name five distinct categories for the chooser evaluation.

Teacher reasoning · W2

Ordinary selection, duration boundaries, invalid records, empty/no-match **input** and ties/order reversal.

Teacher reasoning · W2

Use ordinary eligible selection, duration boundaries, schema-invalid records, empty or no-match **input**, and tied scores with reversed order. Each category checks a different requirement.

Include concrete **input**s and expected reports; five copies of one ordinary case do not provide this **coverage**.

W3 Repair reporting

Workbook question · W3

A learner drops two failed cases and reports 8/8. What should be restored?

Teacher reasoning · W3

Report all 10 executed cases with 8 passes and 2 failures, category details and the tested version.

W4 Evaluate independently

Workbook question · W4

Given ordinary 3/3, boundary 1/2, invalid 2/2 and tie 0/1, compute overall results and prioritize issues.

Teacher reasoning · W4

6/8 pass, 2 fail; boundary has one failure and ties fail the only case. Report both and avoid treating 75 percent as production accuracy.

W5 Protect final evidence

Workbook question · W5

You fix the tie bug using the final tie case. What should the next evaluation report say?

Teacher reasoning · W5

The old case is now a regression/development case. Document the fix and use fresh reserved cases for a new independent final check.

Assess independent transfer

Independent question · withhold the key

Selected case: Given ordinary 3/3, boundary 1/2, invalid 2/2 and tie 0/1, compute overall results and prioritize issues.

Selected case answer

Teacher key · Selected case answer

6/8 pass, 2 fail; boundary has one failure and ties fail the only case. Report both and avoid treating 75 percent as production accuracy.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

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Connect and extend learning

> Connect Test Diverse Users to Transparent Book Recommender, adapt support and plan a later changed-case check.

Transparent Book Recommender

Use the same test record across ordinary, boundary, missing-data, and access-needs cases.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 35: Improve and Challenge. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Use a category-by-case grid. Extension: add a new case that tests a different failure **mechanism** rather than duplicating a passing case.

Independent W4 **evidence**: 6/8 pass, 2 fail; boundary has one failure and ties fail the only case. Report both and avoid treating 75 percent as production accuracy.

Record whether the learner needed no help, a hint or explanation. Fresh retry: ordinary 2/2, empty 0/1, boundary 2/2 gives 4/5 overall; the empty case still fails. Require the failure record and a fresh-case plan.

Next week: Improving without hiding failures. Reuse this week's explicit contracts and **evidence**.

Plan the delayed return

Suggested return points in this level: ai-2/35. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Build a recommender evaluation matrix with routine, boundary, missing, invalid and tie cases, then add separately planned user-condition **coverage**.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Test Diverse Users lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Evaluate across varied cases

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

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>
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>
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> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Name five distinct categories for the chooser evaluation.

Supported practice answer

Reasoned practice answer: Ordinary selection, duration boundaries, invalid records, empty/no-match **input** and ties/order reversal.

Check every learner before discussion: Use the worked case for Evaluating varied cases. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: For a rule-based chooser, evaluate conformance to its policy separately from whether users find suggestions useful. Neither a pass fraction nor an invented satisfaction score proves educational benefit.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Someone reports only the ordinary slice. Require a **prediction** and a reason before comparing. Expected result: The claim becomes 4/4 on ordinary cases, not 100% across all tests.

Reason: Eight nonordinary cases and their failures have been omitted.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

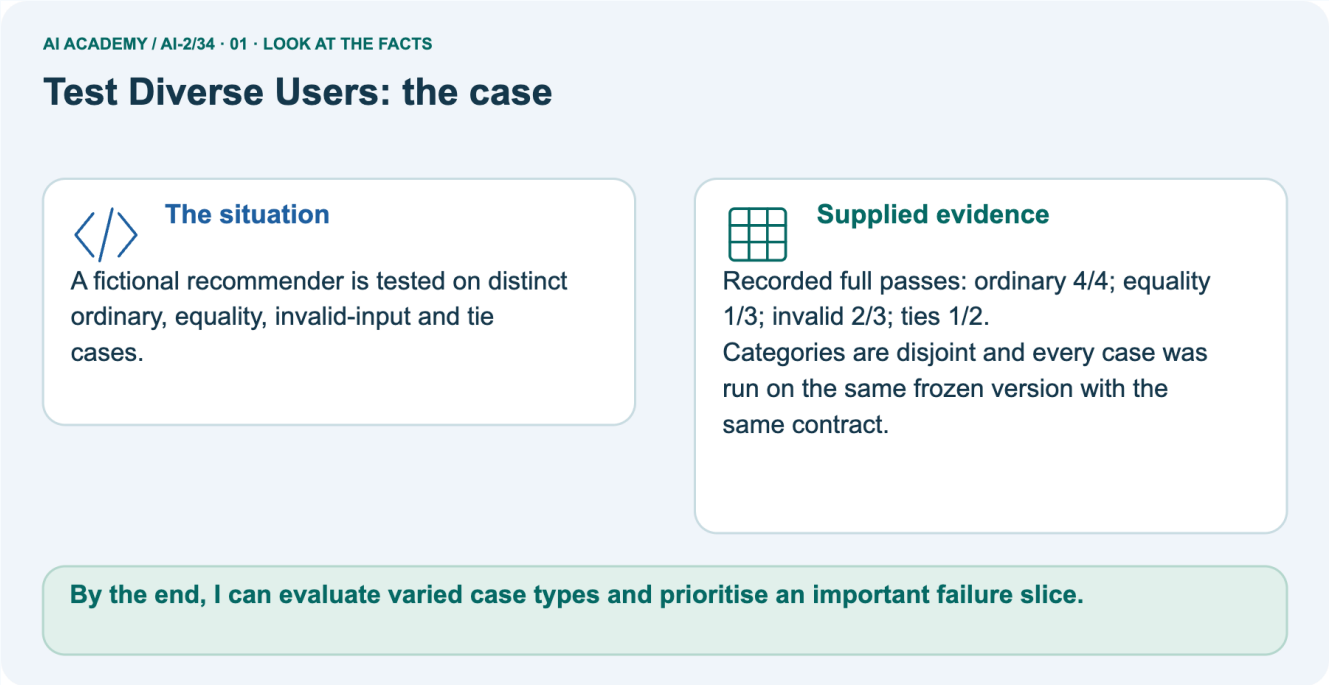
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Test Diverse Users graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Test Diverse Users: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional recommender is tested on distinct ordinary, equality, invalid-**input** and tie cases.
Supplied facts: Recorded full passes: ordinary 4/4; equality 1/3; invalid 2/3; ties 1/2.

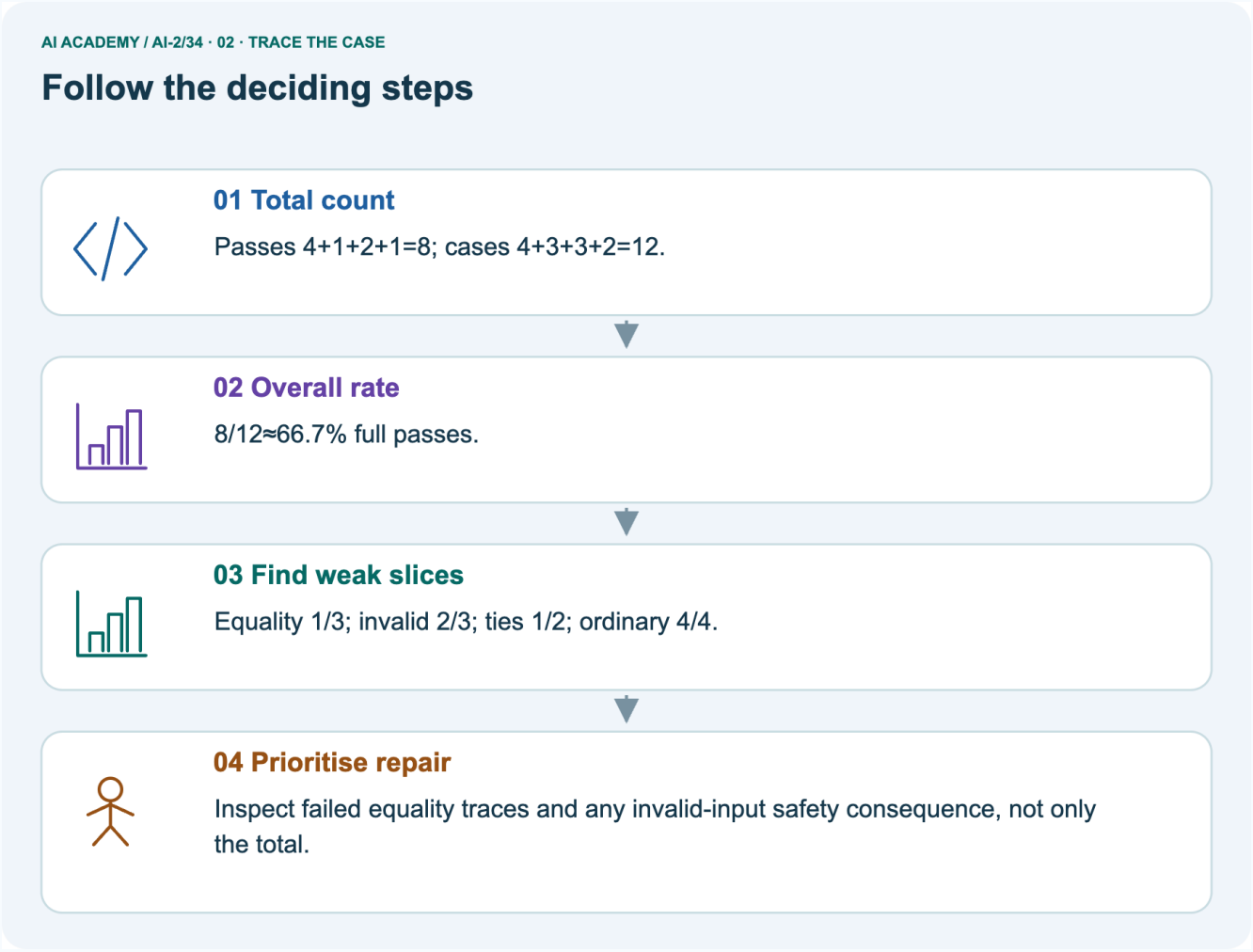
Categories are disjoint and every case was run on the same frozen version with the same contract.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

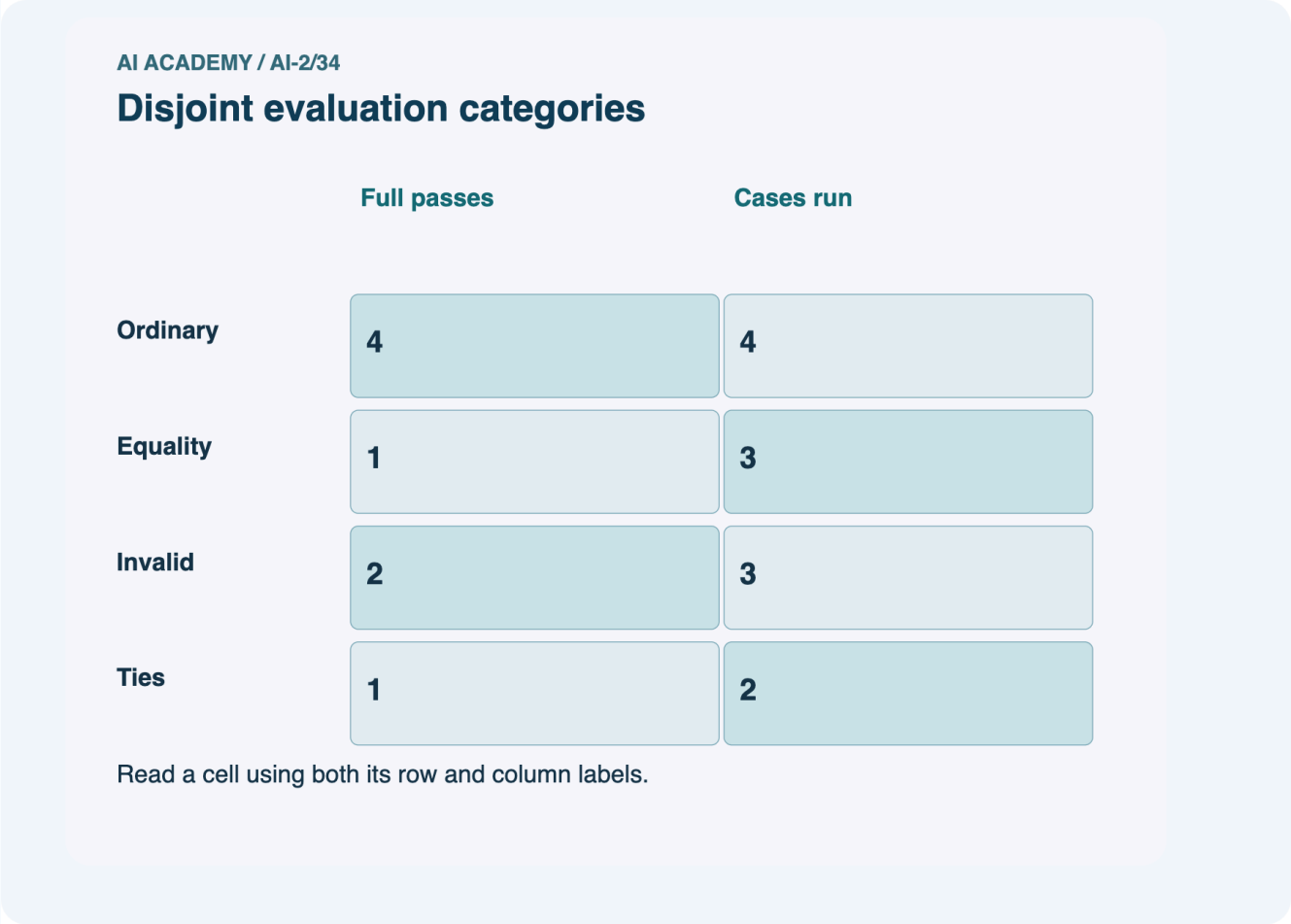
Total count: Passes 4+1+2+1=8; cases 4+3+3+2=12. → Overall rate: 8/12≈66.7% full passes. → Find weak slices: Equality 1/3; invalid 2/3; ties 1/2; ordinary 4/4. → Prioritise repair: Inspect failed equality traces and any invalid-**input** safety consequence, not only the total.

> Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Disjoint evaluation categories



Follow the labelled picture with the matching explanation. Open the vector diagram

Disjoint evaluation categories. Ordinary: Full passes = 4, Cases run = 4. Equality: Full passes = 1, Cases run = 3. Invalid: Full passes = 2, Cases run = 3. Ties: Full passes = 1, Cases run = 2.

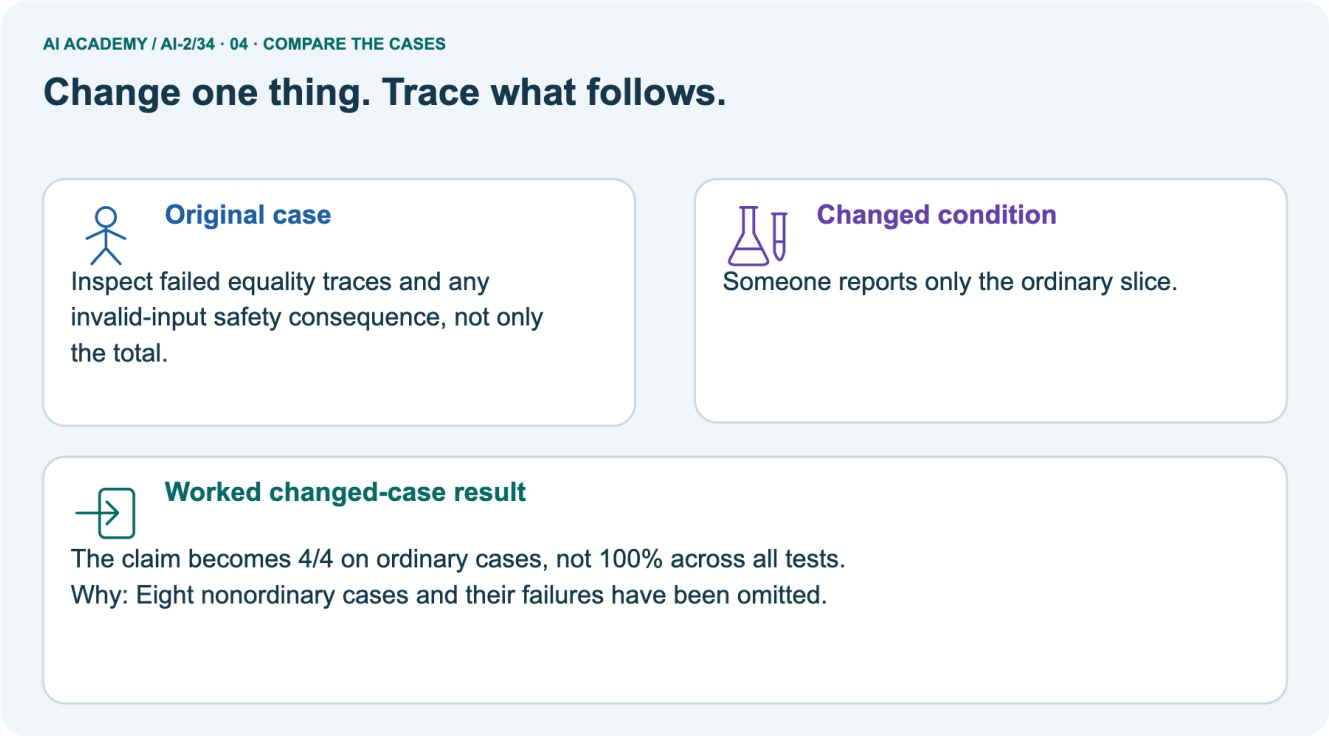
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Inspect failed equality traces and any invalid-**input** safety consequence, not only the total. Changed condition: Someone reports only the ordinary slice.

Expected result: The claim becomes 4/4 on ordinary cases, not 100% across all tests. Reason: Eight nonordinary cases and their failures have been omitted.

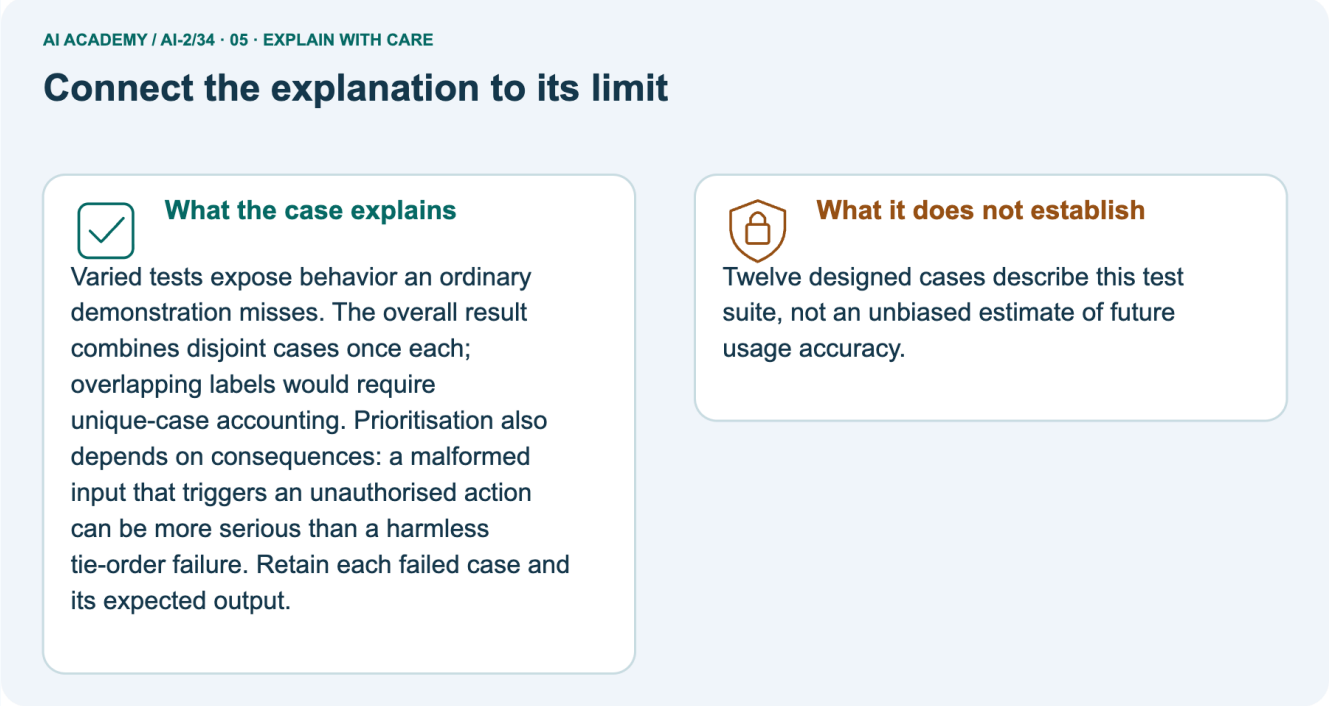
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Eight nonordinary cases and their failures have been omitted.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Varied tests expose behavior an ordinary demonstration misses. The overall result combines disjoint cases once each; overlapping labels would require unique-case accounting.

Prioritisation also depends on consequences: a malformed **input** that triggers an unauthorised action can be more serious than a harmless tie-order failure. Retain each failed case and its expected **output**.

Limit: Twelve designed cases describe this test suite, not an unbiased estimate of future usage accuracy.

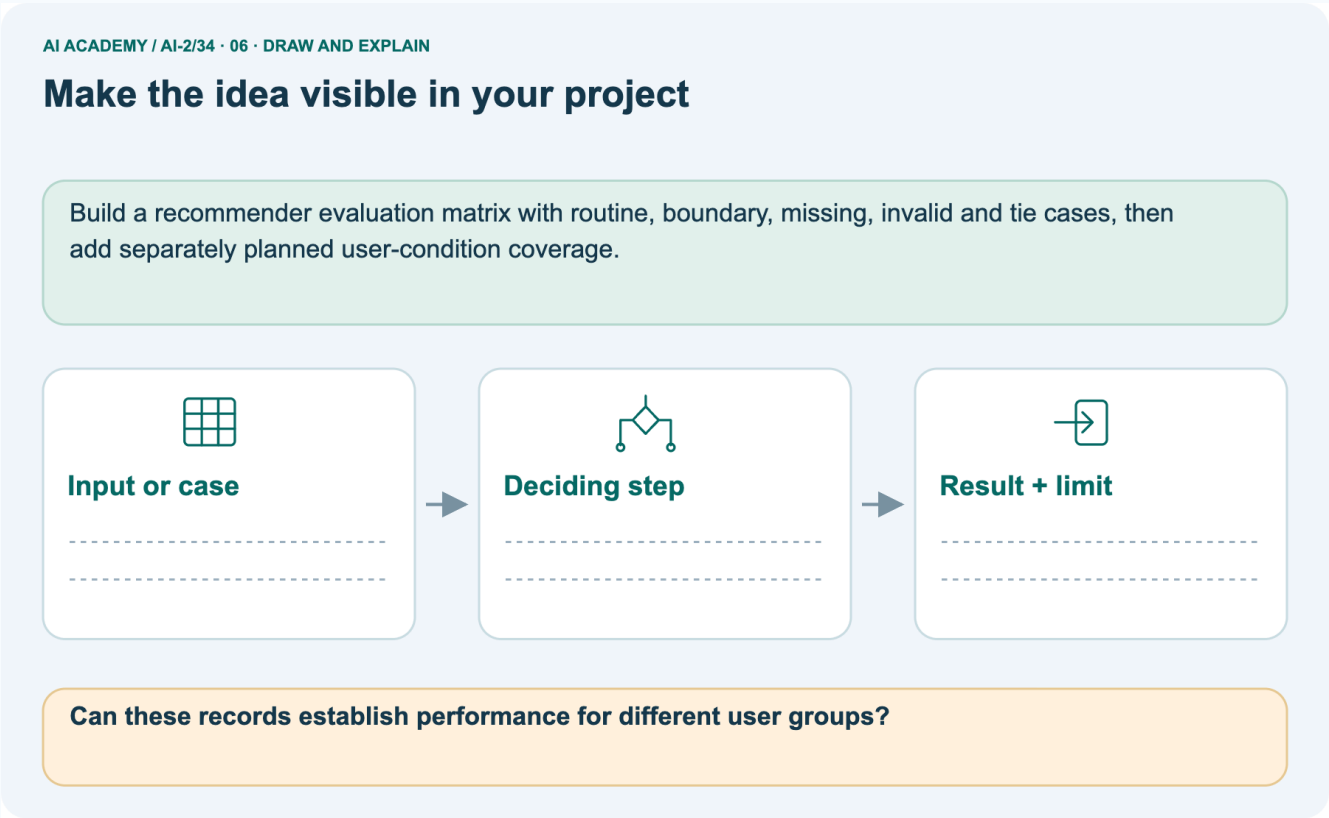
> Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Twelve designed cases describe this test suite, not an unbiased estimate of future usage accuracy.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Build a recommender evaluation matrix with routine, boundary, missing, invalid and tie cases, then add separately planned user-condition **coverage**. Draw your **input**, deciding step, result and limit.

Supported question: Can these records establish performance for different user groups?

> Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

> Supported practice key: No. They describe **input**-case types. No user-group membership or group-specific outcomes were supplied.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.